

## SHANNON, Ireland

WMO: 039620

Lat: 52.700N

Lon: 8.917W

Elev: 14

StdP: 101.16

Time Zone: 1.00 (IRL)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |             |            |            | Coldest Month WS/MCDB |            |             |            | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|------------|-------------|------------|------------|-----------------------|------------|-------------|------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |            | 99%         |            |            | 0.4%                  |            | 1%          |            |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP          | HR         | MCDB       | WS                    | MCDB       | WS          | MCDB       | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>-2.2 | (c)<br>-0.6 | (d)<br>-4.2                   | (e)<br>2.7 | (f)<br>0.1 | (g)<br>-2.7 | (h)<br>3.0 | (i)<br>1.1 | (j)<br>16.0           | (k)<br>8.6 | (l)<br>14.4 | (m)<br>8.8 | (n)<br>2.0            | (o)<br>30 | (p)<br>0.633 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>6.3             | (c)<br>23.5     | (d)<br>17.5 | (e)<br>21.6 | (f)<br>16.7 | (g)<br>20.1 | (h)<br>16.0 | (i)<br>18.6         | (j)<br>21.8 | (k)<br>17.6 | (l)<br>20.2 | (m)<br>16.9 | (n)<br>19.2 | (o)<br>3.9           | (p)<br>120 |

| Dehumidification DP/MCDB and HR |             |             |             |             |             |             |             |             | Enthalpy/MCDB |             |             |             |             |             | Extreme<br>Max WB |
|---------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|-------------|-------------|-------------------|
| 0.4%                            |             |             | 1%          |             |             | 2%          |             |             | 0.4%          |             | 1%          |             | 2%          |             |                   |
| DP                              | HR          | MCDB        | DP          | HR          | MCDB        | DP          | HR          | MCDB        | Enth          | MCDB        | Enth        | MCDB        | Enth        | MCDB        |                   |
| (a)<br>17.3                     | (b)<br>12.4 | (c)<br>19.9 | (d)<br>16.5 | (e)<br>11.8 | (f)<br>19.0 | (g)<br>15.8 | (h)<br>11.2 | (i)<br>18.2 | (j)<br>52.5   | (k)<br>21.7 | (l)<br>49.6 | (m)<br>20.3 | (n)<br>47.3 | (o)<br>19.2 | (p)<br>23.6       |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 12.8              | 11.2 | 9.9 | -3.9                       | 26.4 | 2.0                | 2.3 | -5.4                                               | 28.1 | -6.5       | 29.5 | -7.6       | 30.8 | -9.0       | 32.5 |
|                   |      |     | -4.4                       | 20.0 | 1.9                | 1.3 | -5.7                                               | 20.9 | -6.9       | 21.7 | -7.9       | 22.4 | -9.3       | 23.3 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.8  | 6.3   | 6.5   | 7.6   | 9.6   | 12.1  | 14.6  | 16.1  | 15.9  | 14.3  | 11.5  | 8.5   | 6.6  | (6)  |
| (7)  |                                                                                  | DBStd       | 4.38  | 2.89  | 2.58  | 2.52  | 2.39  | 2.42  | 2.25  | 1.93  | 1.83  | 2.03  | 2.46  | 2.81  | 3.23 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 521   | 118   | 99    | 80    | 34    | 7     | 0     | 0     | 0     | 1     | 13    | 61    | 109  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 2753  | 374   | 331   | 333   | 262   | 194   | 114   | 73    | 78    | 123   | 211   | 296   | 363  | (9)  |
| (10) |                                                                                  | CDD10.0     | 825   | 2     | 2     | 6     | 23    | 71    | 138   | 190   | 184   | 128   | 61    | 14    | 4    | (10) |
| (11) |                                                                                  | CDD18.3     | 13    | 0     | 0     | 0     | 0     | 0     | 3     | 6     | 4     | 1     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 72    | 0     | 0     | 0     | 0     | 4     | 26    | 27    | 14    | 1     | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 11    | 0     | 0     | 0     | 0     | 0     | 5     | 4     | 2     | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 4.7   | 5.2   | 5.2   | 4.9   | 4.8   | 4.7   | 4.4   | 4.2   | 4.3   | 4.4   | 4.7   | 4.6   | 5.0  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 968   | 101   | 76    | 72    | 61    | 63    | 66    | 66    | 83    | 81    | 92    | 101   | 107  | (15) |
| (16) |                                                                                  | PrecMax     | 1272  | 221   | 205   | 134   | 113   | 141   | 167   | 134   | 231   | 168   | 176   | 263   | 233  | (16) |
| (17) |                                                                                  | PrecMin     | 709   | 11    | 1     | 0     | 11    | 10    | 11    | 13    | 4     | 9     | 0     | 43    | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 130   | 47    | 46    | 31    | 27    | 31    | 33    | 29    | 42    | 36    | 39    | 41    | 49   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 13.1  | 13.5  | 16.1  | 19.2  | 23.5  | 26.6  | 26.8  | 25.2  | 22.2  | 18.6  | 15.5  | 14.2 | (19) |
| (20) |                                                                                  |             | MCWB  | 11.5  | 11.4  | 11.4  | 13.5  | 16.7  | 18.6  | 18.7  | 19.7  | 17.6  | 15.6  | 13.8  | 12.7 | (20) |
| (21) |                                                                                  | 2%          | DB    | 12.0  | 12.2  | 13.6  | 16.6  | 20.5  | 22.8  | 23.1  | 22.0  | 20.3  | 17.0  | 14.3  | 12.9 | (21) |
| (22) |                                                                                  |             | MCWB  | 10.7  | 10.5  | 10.5  | 11.9  | 15.2  | 16.3  | 17.7  | 17.5  | 16.3  | 14.6  | 12.8  | 11.7 | (22) |
| (23) |                                                                                  | 5%          | DB    | 11.2  | 11.4  | 12.4  | 15.0  | 18.2  | 20.6  | 21.1  | 20.3  | 18.9  | 16.1  | 13.3  | 11.9 | (23) |
| (24) |                                                                                  |             | MCWB  | 10.0  | 9.9   | 10.0  | 11.3  | 13.5  | 15.5  | 16.9  | 16.6  | 15.5  | 14.1  | 12.1  | 10.7 | (24) |
| (25) |                                                                                  | 10%         | DB    | 10.4  | 10.6  | 11.5  | 13.6  | 16.3  | 18.7  | 19.6  | 19.1  | 17.8  | 15.2  | 12.5  | 11.1 | (25) |
| (26) |                                                                                  |             | MCWB  | 9.4   | 9.2   | 9.5   | 10.5  | 12.5  | 14.6  | 16.1  | 15.9  | 15.0  | 13.5  | 11.3  | 10.0 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 11.8  | 11.8  | 12.1  | 14.2  | 17.3  | 19.0  | 20.1  | 20.1  | 18.5  | 16.3  | 14.2  | 13.0 | (27) |
| (28) |                                                                                  |             | MCDWB | 12.7  | 12.9  | 14.2  | 17.8  | 22.5  | 24.7  | 25.0  | 24.5  | 21.2  | 17.6  | 15.1  | 13.9 | (28) |
| (29) |                                                                                  | 2%          | WB    | 10.9  | 10.9  | 11.2  | 12.7  | 15.6  | 17.2  | 18.6  | 18.5  | 17.2  | 15.2  | 13.1  | 11.8 | (29) |
| (30) |                                                                                  |             | MCDWB | 11.7  | 11.8  | 12.8  | 15.4  | 19.6  | 21.6  | 21.7  | 21.1  | 19.1  | 16.4  | 14.0  | 12.7 | (30) |
| (31) |                                                                                  | 5%          | WB    | 10.2  | 10.2  | 10.6  | 11.9  | 14.2  | 16.1  | 17.5  | 17.5  | 16.4  | 14.5  | 12.3  | 10.9 | (31) |
| (32) |                                                                                  |             | MCDWB | 11.0  | 11.2  | 12.0  | 14.1  | 17.0  | 19.5  | 20.0  | 19.5  | 18.1  | 15.7  | 13.1  | 11.7 | (32) |
| (33) |                                                                                  | 10%         | WB    | 9.5   | 9.4   | 9.8   | 10.9  | 13.2  | 15.1  | 16.7  | 16.7  | 15.5  | 13.7  | 11.5  | 10.1 | (33) |
| (34) |                                                                                  |             | MCDWB | 10.4  | 10.5  | 11.3  | 13.1  | 15.5  | 18.1  | 18.9  | 18.5  | 17.2  | 14.9  | 12.4  | 11.0 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 4.8   | 5.5   | 6.4   | 7.2   | 7.2   | 7.0   | 6.3   | 6.2   | 6.4   | 5.7   | 5.1   | 4.8  | (35) |
| (36) |                                                                                  |             | MCDBR | 5.0   | 5.5   | 7.3   | 8.9   | 10.3  | 10.5  | 8.9   | 7.9   | 7.8   | 5.8   | 4.9   | 5.1  | (36) |
| (37) |                                                                                  |             | MCWBR | 4.7   | 4.7   | 5.0   | 5.4   | 5.7   | 5.4   | 4.9   | 4.6   | 4.7   | 4.3   | 4.4   | 4.8  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 5.0   | 5.2   | 6.4   | 7.7   | 9.1   | 9.1   | 7.8   | 6.8   | 6.5   | 5.4   | 4.8   | 5.1  | (38) |
| (39) |                                                                                  |             | MCWBR | 4.8   | 4.8   | 5.0   | 5.2   | 5.5   | 5.2   | 4.8   | 4.6   | 4.9   | 4.6   | 4.5   | 4.9  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.341 | 0.349 | 0.371 | 0.380 | 0.393 | 0.377 | 0.382 | 0.364 | 0.360 | 0.346 | 0.334 | 0.331 | (40) |      |
| (41) |                                                                                  | taud        | 2.245 | 2.307 | 2.276 | 2.286 | 2.283 | 2.363 | 2.358 | 2.417 | 2.423 | 2.415 | 2.304 | 2.215 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 618   | 748   | 809   | 852   | 857   | 875   | 862   | 860   | 818   | 747   | 626   | 548   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 67    | 84    | 105   | 118   | 124   | 116   | 114   | 102   | 90    | 75    | 62    | 58    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.72  | 1.40  | 2.51  | 3.92  | 4.78  | 5.05  | 4.54  | 3.87  | 2.85  | 1.68  | 0.87  | 0.54  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.07  | 0.12  | 0.24  | 0.34  | 0.28  | 0.43  | 0.36  | 0.27  | 0.20  | 0.14  | 0.07  | 0.07  | (45) |      |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |         |         |         |     |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|-----|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |     |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     | (m) |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +82     | N/A     | N/A |
| (47) | Regional (18 neighbors) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A |

Nomenclature: See separate page